LAB4: Subnetting and Supernetting using Cisco Packet Tracer

OBJECTIVE:

- To design and simulate a subnetting and supernetting using Cisco Packet Tracer.
- To verify communication between computers using ping.

TOOLS USED

- Cisco Packet Tracer

THEORY:

SUBNETTING:

Subnetting is a network design technique that divides a larger IP network into smaller, more manageable segments called subnets. This process allows for efficient utilization of IP addresses, minimizing wastage and ensuring that each subnet has an appropriate number of addresses for its devices. Subnetting also helps reduce network congestion, enhances security by isolating different network segments, and simplifies network management by organizing devices logically.

Additionally, subnetting supports better routing control, efficient broadcast management, and scalability, making it essential for both small and large enterprise networks.

SUPERNETTING:

Supernetting combines multiple smaller networks into a larger, single network. The main purpose of supernetting is to reduce the number of routing table entries, simplify routing, and improve overall network efficiency. This reduces the complexity of routing and helps conserve memory and processing resources in routers. Supernetting is commonly used in large enterprise networks and on the Internet.

NETWORK TOPOLOGY:

Switch :

A switch is an advanced networking device used to connect devices within a LAN. It operates at Data Link Layer of the OSI model. Unlike a hub a switch forwards data only to intended destination by using MAC addresses which significantly reduces collisions and improves network performance.

Router:

A router is Network Layer device used to connect different IP networks and enable communication between them. Unlike switches which operates within a single network, routers forward data packets between multiple networks based on IP addresses. When a device needs to communicate with a device in another network, the data is first sent to the default gateway, and the router forwards the packet to the destination network.

NETWORK DESIGN:

Base network 192.168.1.0/24 required number of subnets is 4.
Number of IP addresses per subnets = 64

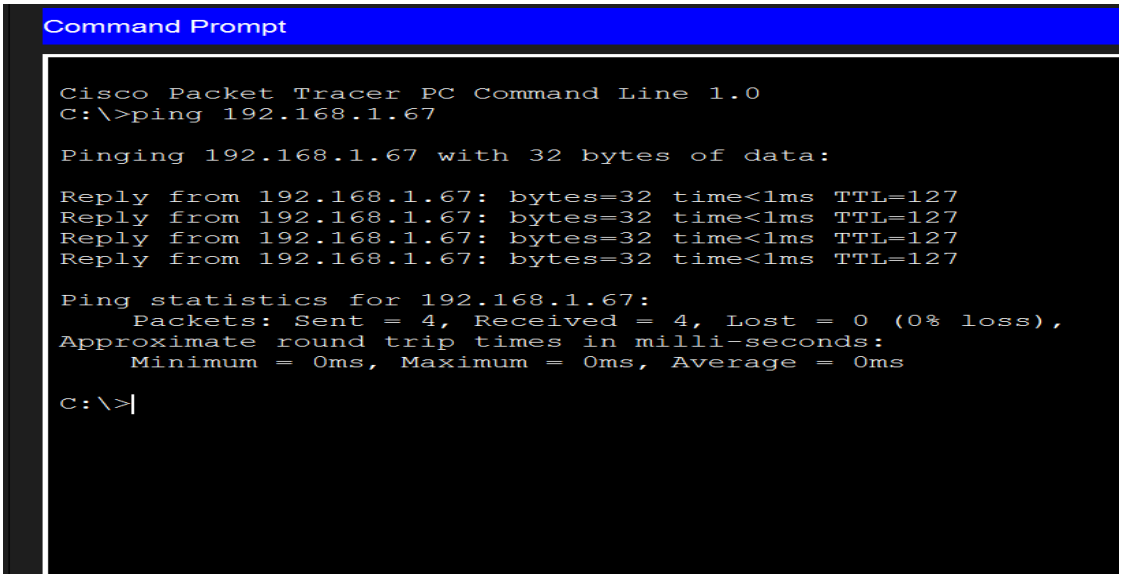
Subnet Table:

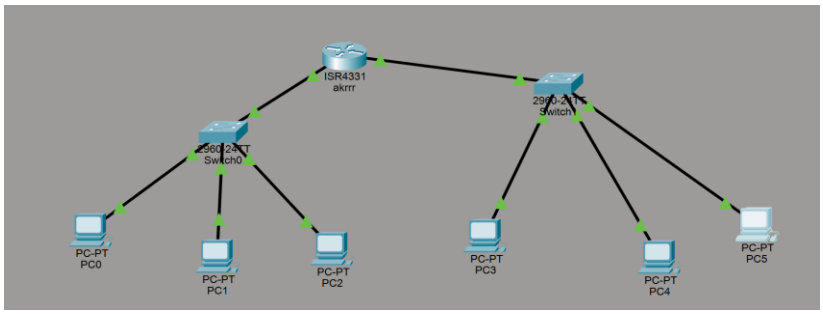
Subnet	Network address	Broadcast Address	1st usable	Last usable
1	192.168.1.0/26	192.168.1.63/26	192.168.1.1/26	192.168.1.62/26
2	192.168.1.64/26	192.168.1.127/26	192.168.1.65/26	192.168.1.126/26
3	192.168.1.128/26	192.168.1.191/26	192.168.1.129/26	192.168.1.190/26
4	192.168.1.192/26	192.168.1.254/26	192.168.1.193/26	192.168.1.253/26

CONFIGURATION and OUTPUTS:

1. Subnetting:

First Ping to check if the connection works.





1. **Supernetting:**
   Ping to check if the connection works

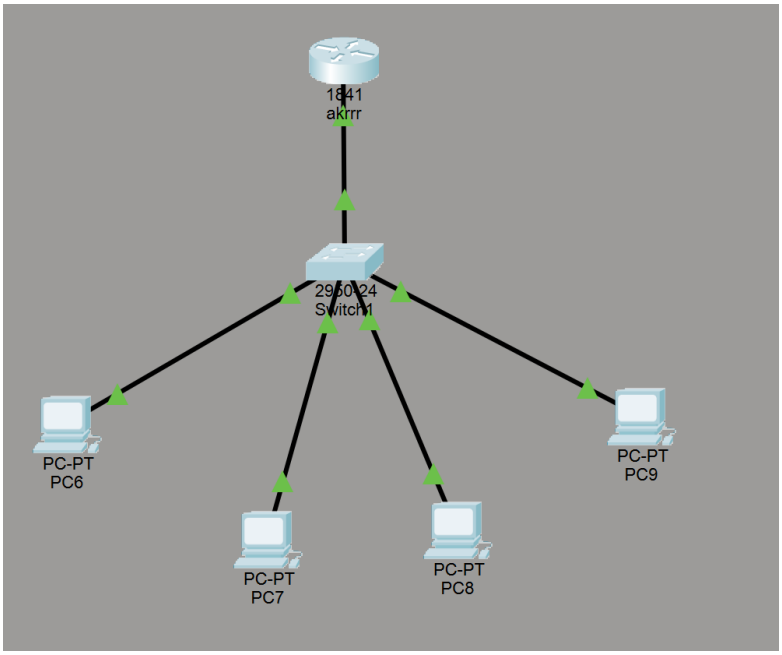
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.10

Pinging 192.168.3.10 with 32 bytes of data:

Reply from 192.168.3.10: bytes=32 time<1ms TTL=128
Reply from 192.168.3.10: bytes=32 time<1ms TTL=128
Reply from 192.168.3.10: bytes=32 time<1ms TTL=128
Reply from 192.168.3.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.3.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```



Discussion and Conclusion:

In this lab, Cisco Packet Tracer was used to perform subnetting and supernetting to better understand IP address management and routing. Subnetting allowed a large network to be split into smaller sub-networks, improving IP address efficiency, reducing network congestion, and making network management easier. Supernetting, on the other hand, merged several smaller networks into a larger network, which helped shrink routing tables and simplify routing.

The configurations were tested in the simulator, and successful packet delivery confirmed that both techniques were implemented correctly.